

AI Usage & Prompt Documentation

Overview

During the development of this project, multiple AI tools were used to accelerate development, improve architecture quality, and automate intelligent workflows.

The project combines:

ChatGPT for backend development, architecture planning, debugging, and documentation

Gemini AI for frontend UI/dashboard improvements

Claude AI (Anthropic) for deployment guidance, troubleshooting, and project validation

Ollama (Llama3) as the runtime AI model for generating SSL renewal tasks dynamically inside the application

These AI systems significantly improved development speed and overall project quality.

AI Tools Used

AI Tool	Purpose
ChatGPT	Backend development, debugging, architecture, reports, documentation
Gemini AI	Frontend UI/UX improvements and dashboard styling
Claude AI	Deployment guidance, Windows troubleshooting, project validation, ngrok setup
Ollama + Llama3	Runtime AI model for SSL renewal task generation

Prompts Given to ChatGPT

◆ Prompt 1 — Project Architecture

Prompt:

Build a Python project that detects SSL certificate expiry dates for public domains, stores results in SQLite, sends Discord alerts, and uses AI to generate renewal tasks.

AI Response:

Suggested modular architecture

Recommended components: scanner.py, database.py, engine.py, dashboard.py, llm.py

Proposed Flask dashboard workflow

Suggested report generation structure

◆ Prompt 2 — SSL Scanner Development

Prompt:

Write Python code to fetch SSL certificate expiry date and calculate remaining days for a public domain.

AI Response:

Generated SSL socket scanning logic

Implemented expiry-date extraction

Added remaining-days calculation

◆ Prompt 3 — Database Integration

Prompt:

Create SQLite database logic for storing SSL monitoring history with status, incidents, renewal tasks, and timestamps.

AI Response:

Designed SQLite schema

Added insert/fetch operations

Fixed schema mismatch errors

◆ Prompt 4 — Flask Dashboard UI

Prompt:

Create a professional Flask dashboard for SSL monitoring with filters, status badges, summary cards, and AI-generated tasks.

AI Response:

Created dashboard structure

Added search filters, summary cards, status badges, auto-refresh support, AI task sections

Gemini AI Usage

Gemini AI was used mainly for frontend enhancement and UI design improvements.

◆ Prompt Given to Gemini

Improve the Flask dashboard UI with modern styling, summary cards, badges, dark theme, responsive tables, and monitoring dashboard appearance.

Gemini Contributions:

Enhanced HTML structure

Improved CSS styling

Designed dashboard cards

Added responsive UI improvements

Improved visual appearance for hackathon presentation

Claude AI

Claude AI was used for complete deployment guidance, Windows-specific troubleshooting, and project requirement validation throughout the entire deployment phase.

◆ Prompt 1 — Project Deployment

Prompt:

How to deploy this SSL Certificate Expiry Watcher project on Windows and get a public link?

Claude Response:

Provided complete step-by-step Windows deployment guide

Explained virtual environment setup
Guided through dependency installation
Recommended ngrok for public URL generation
Provided exact PowerShell commands for Windows

◆ Prompt 2 — ngrok Setup and Troubleshooting

Prompt:

ngrok is not recognized as a command in PowerShell. How do I fix this?

Claude Response:

Diagnosed that ngrok.exe was extracted as a folder instead of a file
Provided exact PowerShell commands to move and fix the file
Explained authtoken configuration steps
Guided through the full ngrok tunnel setup

◆ Prompt 3 — Ollama Integration Fix

Prompt:

LLM Connection Error: HTTP Connection Pool localhost 11434 Read timed out. All domains showing same generic task.

Claude Response:

Diagnosed timeout was set to 30 seconds — too short for llama3
Provided fix to increase timeout to 120 seconds in llm.py
Guided through Ollama installation on Windows
Explained ollama pull llama3 and ollama serve commands

◆ Prompt 4 — Project Requirement Validation

Prompt:

IM-05 SSL Certificate Expiry Watcher — Reads a list of public hostnames, fetches their SSL cert metadata, ranks by days-to-expiry; LLM drafts a renewal task with owner placeholder. Outputs CSV + email-ready Markdown. Does my project satisfy this?

Claude Response:

Validated all 7 required features are fully implemented

Confirmed CSV and Markdown output

Confirmed LLM task generation with owner assignment

Identified 6 bonus features beyond requirements

Confirmed 100% requirement satisfaction

◆ Prompt 5 — Discord Alert Verification

Prompt:

How do I set up Discord alerts and verify they are working in my project?

Claude Response:

Explained Discord webhook creation process

Guided through replacing webhook URL in discord_alert.py

Confirmed alerts fire automatically for WARNING and BROKEN status domains
Verified alert delivery from scan output logs

Ollama AI Model Integration

The project integrates Ollama with the Llama3 model to generate SSL renewal tasks dynamically during runtime.

◆ Runtime Prompt Sent to Ollama

Generate SSL renewal task for:

Domain: wikipedia.org

Days Left: 27

Status: WARNING

Return JSON format with:

- task
- priority
- owner
- action

◆ Example Ollama Response

```
json{  
  "task": "Renew SSL certificate for wikipedia.org",  
  "priority": "P1",  
  "owner": "platform-team",  
  "action": "Generate new certificate and update DNS configuration"  
}
```

AI-Assisted Development Workflow

Problem Statement



ChatGPT Planning & Backend Development



Gemini Frontend UI Enhancement



Claude AI Deployment & Troubleshooting



Ollama AI Task Generation Integration



Testing & Debugging



Dashboard + Reports



Final Integrated System

Impact of AI Assistance

Using AI tools significantly improved:

Development speed

System architecture quality

Frontend UI quality

Debugging efficiency

Documentation quality

Deployment success

Feature completeness

Windows compatibility

Final Outcome

With AI-assisted development, the project successfully achieved:

SSL certificate monitoring

Risk analysis engine

AI-generated renewal tasks (Ollama + Llama3)

Discord alert automation

CSV & Markdown report generation

Flask dashboard visualization

SQLite database integration

Domain ranking by urgency

Public deployment via ngrok

Windows deployment support

Conclusion

This project demonstrates how multiple AI systems can collaboratively accelerate software development:

ChatGPT → Backend + architecture + debugging

Gemini → Frontend UI/UX enhancements

Claude AI → Deployment, troubleshooting, validation

Ollama/Llama3 → Runtime AI task generation

The combination of AI-assisted development and DevOps automation enabled the creation of a production-style SSL observability platform for proactive certificate management.□□□□□□